

Ayano Hiranaka

Curriculum Vitae

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Research Interests

I design AI agents, both embodied and non-embodied, that “co-evolve” with humans through effective communication and collaboration. My research focuses on building **human-AI teams that foster mutual improvement** by adapting to and learning from one another. Recently, I have been working on developing AI-assistants that **resolve human misconceptions** to improve their long-term performance and independence.

Education

- 2024 - **PhD in Computer Science**, *University of Southern California*; GPA: 4.00/4.00
2023 **MS in Mechanical Engineering**, *Stanford University*; GPA: 4.02/4.30
2019 **BS in Mechanical Engineering**, *University of Illinois at Urbana-Champaign*; GPA: 3.98/4.00
Graduation with Highest Honors

Publications

*: denotes equal contribution, †: denotes equal advising

- [paper](#) **Fix the Mind, Not the Move: Interpretable AI Assistance via Knowledge-Gap Localization**
[website](#) Ayano Hiranaka*, Ya-Chuan Hsu*, Stefanos Nikolaidis, Erdem Biyik†, Daniel Seita†
International Conference on Machine Learning (ICML), 2026
- [paper](#) **Active Reward Learning and Iterative Trajectory Improvement from Comparative Language Feedback**
[website](#) Eisuke Hirota*, Zhaojing Yang*, Ayano Hiranaka, Miru Jun, Jeremy Tien, Stuart J. Russell, Anca Dragan, Erdem Biyik
The International Journal of Robotics Research (IJRR), 2025
- [paper](#) **HERO: Human-Feedback-Efficient Reinforcement Learning for Online Diffusion Model Finetuning**
[website](#) Ayano Hiranaka*, Shang-Fu Chen*, Chieh-Hsin Lai*, Dongjun Kim, Naoki Murata, Takashi Shibuya, Wei-Hsiang Liao, Shao-Hua Sun†, Yuki Mitsufuji†
International Conference on Learning Representations (ICLR), 2025
- [paper](#) **NOIR: Neural Signal Operated Intelligent Robots for Everyday Activities**
[website](#) Ruohan Zhang*, Sharon Lee*, Minjune Hwang*, Ayano Hiranaka*, Chen Wang, Wensi Ai, Jin Jie Ryan Tan, Shreya Gupta, Yilun Hao, Gabrael Levine, Ruohan Gao, Anthony Norcia, Li Fei-Fei, Jiajun Wu
Conference on Robot Learning (CoRL), 2023
- [paper](#) **Primitive Skill-based Robot Learning from Human Evaluative Feedback**
[website](#) Minjune Hwang*, Ayano Hiranaka*, Sharon Lee, Chen Wang, Li Fei-Fei, Jiajun Wu, Ruohan Zhang
International Conference on Intelligent Robots and Systems (IROS), 2023
- [paper](#) **A Dual Representation Framework for Robot Learning with Human Guidance**
[website](#) Ruohan Zhang*, Dhruva Bansal*, Yilun Hao*, Ayano Hiranaka, Jialu Gao, Chen Wang, Roberto Martin-Martin, Li Fei-Fei, Jiajun Wu
Conference on Robot Learning (CoRL), 2022
- Best paper award at Aligning Robot Representations with Humans workshop**

Research Experiences

- May 2026 - **Toyota Research Institute, Research Intern at Human Interactive Driving Team**
Cambridge (MA, USA)
- Developing AI agent for driving assistance.
- Dec 2023 - **Sony AI, Research Intern at Deep Generative Modeling Team**
Nov 2024 *Sony AI (Tokyo, Japan)*
- Developed human-feedback-efficient RLHF algorithm for text-to-image diffusion model finetuning
 - Algorithm can train a model for various tasks while also capturing human preference
- Mar 2021 - **Stanford Vision and Learning Lab, Graduate Research Assistant**
Dec 2023 *Stanford University (Stanford CA, USA)*
- Led real robot experiments in multiple human-robot collaboration projects
 - Experience with physical robots, including mobile manipulators (Sawyer, Franka, TIAGo)
 - Experiences in human-in-the-loop robot learning, reinforcement learning, imitation learning, motion planning, brain-robot-interface
- Sep 2019 - **Machine Tool Systems Research Lab, Undergraduate Researcher**
Dec 2019 *University of Illinois at Urbana-Champaign (Champaign IL, USA)*
- Investigated effect of atomization-based cutting fluid (ACF) spray angle and distance on tool life during micro-drilling operations
- Sep 2018 - **Mehta Research Group, Undergraduate Researcher**
Jun 2019 *University of Illinois at Urbana-Champaign (Champaign IL, USA)*
- Developed adaptive particle filter algorithm for real-time piano note pitch identification

Teaching Experiences

- 2022 **ENGR 110/210: Perspectives in Assistive Technology**, *Stanford University*
Graduate Teaching Assistant (Winter)
- 2021 **ME 161: Dynamic Systems, Vibrations and Control**, *Stanford University*
Graduate Teaching Assistant (Fall)

Honors and Awards

- May 2026 **ICML 2026 Gold Reviewer**
- May 2020 **Bronze Tablet Recipient**, *University of Illinois at Urbana-Champaign*
- Dec 2019 **Graduation with Highest Honors**, *University of Illinois at Urbana-Champaign*

Services

Reviewing: I serve as a reviewer for ICLR, ICML, CoRL, ICRA, and workshops at RSS

Mentoring: Organizing and serving as mentor for USC CS Undergraduate Mentorship Program that prepare undergraduate students for research and career in CS

Skills

AI: Human-in-the-loop learning, human-AI-teaming, HRI/HRC, shared autonomy, hierarchical learning, RL, IL, diffusion models, representation learning, AI teaching

Hardwares: Franka, Sawyer, TIAGo, WidowX

Robotics: ROS, controls, mobile manipulation, task and motion planning, camera calibration

Languages: Japanese (speak, read, write fluently)